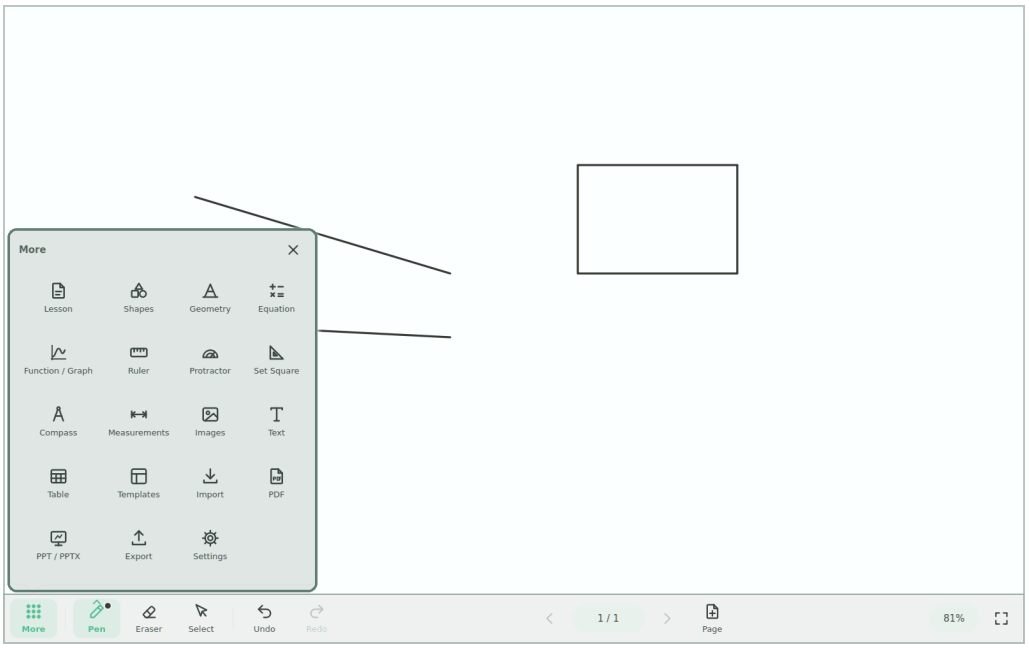
ClassBoard

A smart digital board for classrooms – simple guide

In one line: ClassBoard turns a big touch screen in a classroom into a smart blackboard. You write with your finger or a pen, and the board helps you draw neat shapes, make maths equations, plot graphs and save the whole lesson – all without internet.



The board: writing area on top, buttons (the “ribbon”) at the bottom, the MORE menu open.

Made with	C++17 and Qt 5.15
Runs on	Windows 10 / 11 (also builds on Linux)
Internet needed?	No – works fully offline
Licence	MIT (open source, free to use)
Code	github.com/An0nym0us-glitch/Smart-board
Website	an0nym0us-glitch.github.io/Smart-board-website

What’s inside this guide

1. What is ClassBoard?	5. Geometry, equations, graphs
2. How to use the board	6. Pages, saving, exporting
3. The tools	7. How it’s built inside
4. The smart (AI) features	8. The ClassBoard website
9. Questions people may ask	10. Simple words (glossary)

1. What is ClassBoard?

Many schools now have a big interactive display instead of (or next to) a blackboard. ClassBoard is the software that runs on that display. When it opens, the whole screen becomes a green blackboard and the pen is already selected, so a teacher can start writing straight away.

Why it's useful

- **Feels like a real board** – write with a finger, a stylus pen or a mouse.
- **Many people can write at once** – up to 10 fingers, each drawing its own line, so several students can come up together.
- **Makes messy drawings neat** – a rough circle becomes a perfect circle.
- **Built for maths and science** – ruler, protractor, compass, graphs, equations and exact measurements.
- **Lessons are saved** – the teacher can save, reopen, and export to PDF or PowerPoint for students.
- **No internet needed** – everything runs on the computer, so it works in any classroom.

> **Say it like this:** *"ClassBoard is a free digital blackboard for smart classroom screens. You write on it like a normal board, but it can clean up your drawings, read handwritten maths, draw graphs, and save the whole lesson as a PDF."*

2. How to use the board

The buttons along the bottom

All the main buttons sit in one strip at the bottom of the screen, called the **ribbon**. Everything else opens in small pop-up panels (called **popovers**) right next to the button you tapped, so you never leave the board.

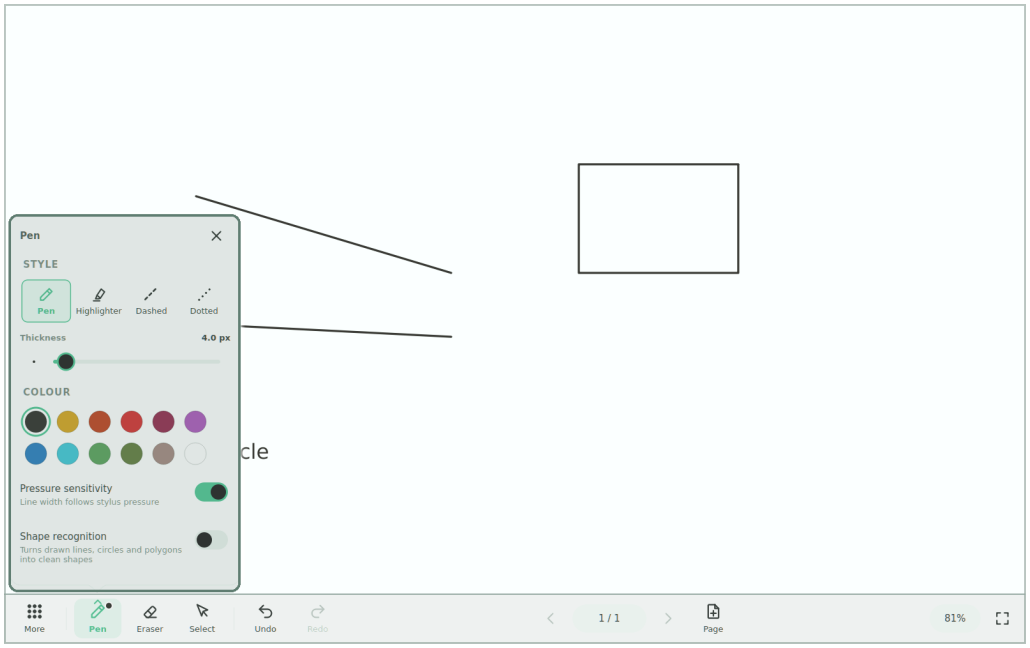
Button	What it does
MORE	Opens every other tool: shapes, geometry, equations, graphs, ruler, images, text, tables, templates, import/export, settings.
PEN / ERASER / SELECT	Switch tool. Tap the same button again to see its options (colour, thickness…).
UNDO / REDO	Go back or forward one step. There is only one undo list for everything.
FILE / INSERT / EDIT	New, open, save · add image/shape/graph · copy, paste, delete.
‹ 3 / 12 ›	Shows the page number. Tap it to see all pages as small pictures.
NEW PAGE / PAGE	Add a blank page · page menu (duplicate, clear, delete, size, background).
− 81% +	Zoom out / in. Tap the number for more view options. Also full screen.

Touch, pen and mouse

Action	Touch	Pen	Mouse
Draw	One finger (each finger draws its own line)	Pen tip	Left button
Move the board	Two fingers	–	Right / middle drag
Zoom	Pinch with two fingers	–	Mouse wheel
Erase	Three fingers together or flat hand – wipe	Back end of pen	–

Clever touch tricks
• **Palm wipe:** put three fingers side by side (or your flat hand) and wipe like a duster. It works in any tool and is one undo step.
• **Hand resting on the screen?** While the pen is touching or hovering, finger touches are ignored, so your hand doesn't draw by mistake.
• **Young kids zooming by accident?** Turn on "Every finger draws" in Settings – then two-finger zoom is off.

3. The tools



The Pen options: style, colour, thickness, pressure and shape recognition.

Tool	In simple words
Pen	Normal pen, highlighter, dashed and dotted lines. Choose colour and thickness. Pen pressure makes lines thicker when you press harder.
Magic Highlighter	For pointing at something while explaining. The mark fades away by itself after about 5 seconds and is never saved.

Tool	In simple words
Eraser	**Stroke** – removes a whole line. **Object** – removes anything you touch. **Area** – rubs out only what's under it, like a real duster (you can erase just one side of a triangle).
Select	Tap, drag a box, or draw a loop (lasso) around things. Then move, resize, rotate, recolour, copy or delete them.
Shapes	Line, arrow, rectangle, circle, ellipse, triangle, diamond, hexagon, polygon and more – with no fill, light fill or solid fill.
Text, Images, Tables	Type text with bold/italic, drop in pictures, or add a table.

4. The smart (AI) features

ClassBoard has three "smart" features. The important thing to explain: **they all run on the computer itself, with no internet and no big AI model**. They use clever maths rules, so they are fast and give the same result every time – which is good in a classroom.

A. Shape recognition – messy drawing → neat shape

Turn it on in the Pen options. Draw a rough circle, line, rectangle or triangle, and it snaps into a clean shape.

How it works (simple)
1. When you lift the pen, the program looks at all the points of your line.
2. It tries to **fit** different shapes to those points – a straight line, a circle, an ellipse, a triangle, a rectangle, up to a hexagon.
3. It checks how close each shape fits and gives it a **confidence** score.
4. If one shape fits well, it replaces your ink with that shape. If nothing fits well, your drawing stays as it is.

This is called geometric fitting – no training data, just geometry.

B. ✦ Magic Equation Maker – handwriting → typed equation

Write a maths expression with the pen (for example $x^2 + 1$ or ½), select it, and tap the magic button. ClassBoard reads your handwriting and shows a preview. The teacher then chooses:

Button	What happens
Accept	Your handwriting is replaced by a neat, editable equation. One undo brings the handwriting back.
Edit	Fix the equation first, then insert it.
Cancel	Nothing changes.
Try again	Shown if it couldn't read it – it gives its best guess.

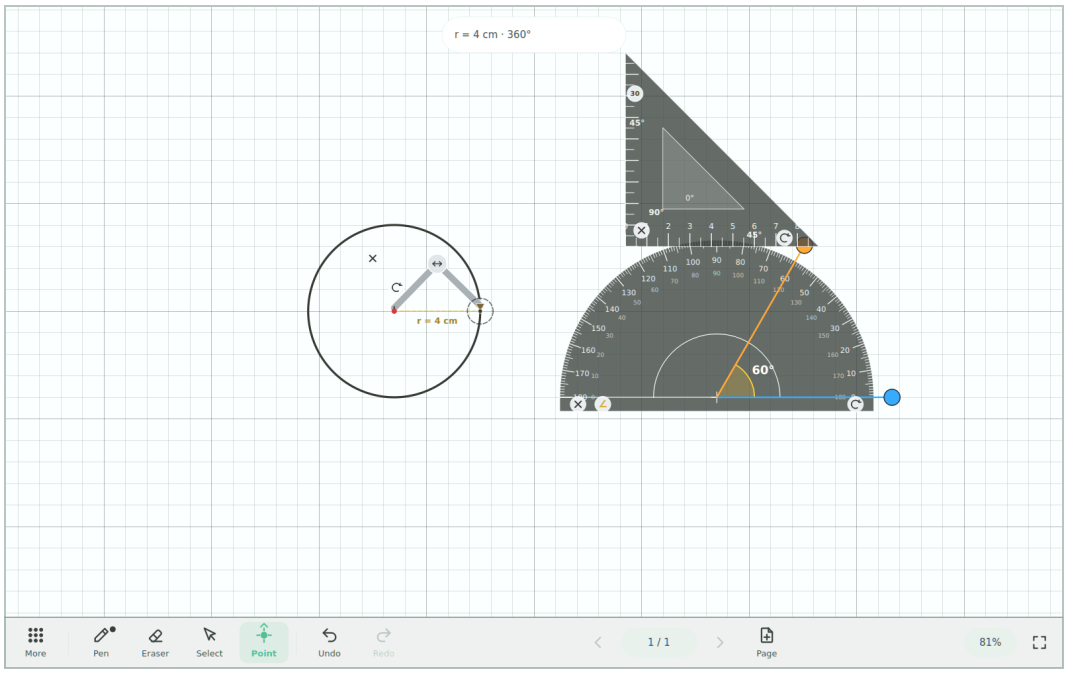
How it reads handwriting (simple)
1. **Group strokes into symbols** – strokes that cross or overlap (like the two lines of “+” or “x”) are one symbol.
2. **Find flat lines** – a flat line with writing above and below is a **fraction bar**; two flat lines close together are **=**.
3. **Recognise each symbol** – each symbol is turned into a “cloud” of points and compared with built-in example shapes of 0–9, a, b, c, n, x, y, +, (and). The closest match wins. (This method is called the **$P point-cloud recogniser**; it doesn’t care in which order or direction you drew the strokes.)
4. **Work out the layout** – a small symbol raised up next to another is a **power** (x^2).
5. **Write LaTeX** – the result becomes maths code like `x^{2}+1` or `\frac{1}{2}`, which is drawn as a proper equation.
Safety rule: if it is unsure about even one symbol, it says so instead of guessing. It never changes the teacher’s writing without permission.

C. Magic Highlighter – marks that disappear

Draw to point at something; the mark stays about 5 seconds, then fades. These marks live on a separate temporary layer, so they are never saved, exported or added to undo.

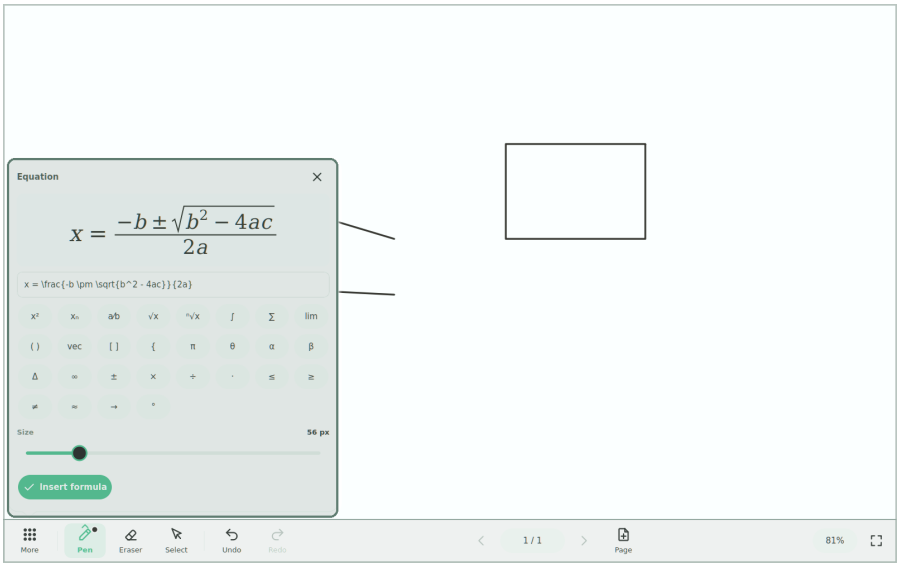
> **Say it like this:** *“The smart features don’t need internet. Shape recognition fits real geometric shapes to your drawing, and the equation maker compares each handwritten symbol to stored examples, then builds the equation. If it isn’t sure, it asks the teacher instead of guessing.”*

5. Geometry, equations and graphs



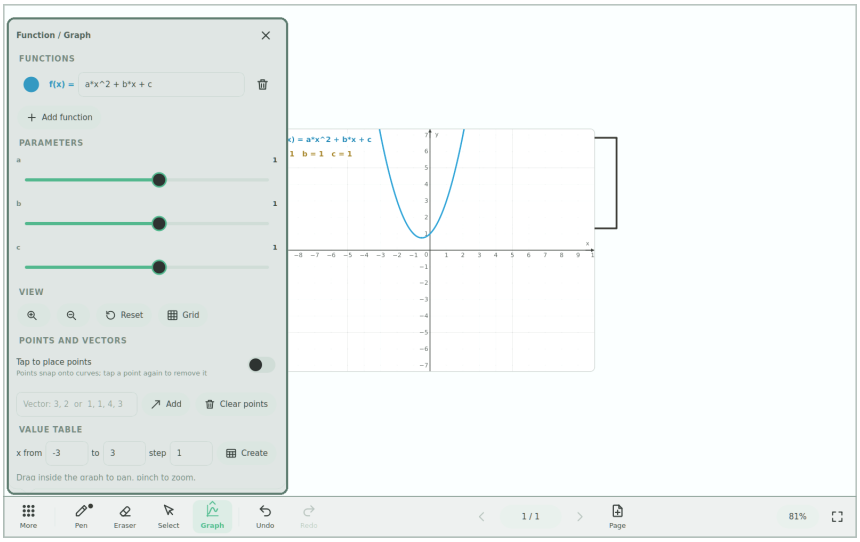
Ruler, protractor, set square and compass lying on the board.

Feature	What you can do
Instruments	**Ruler / set square:** draw near an edge and the line follows it exactly, with the length shown. **Protractor:** drag two knobs to measure an angle and stamp it. **Compass:** drag the pencil to draw arcs and circles.
Measurements	Distance, angle, slope and area – the numbers update live when you move the points.
Constructions	Points (A, B, C… with coordinates), segments, lines, rays and vectors. They snap to points and to the grid.
Exact values	Select a line or shape and type the exact length, angle or slope (e.g. 60° in one tap).
Scale	Set e.g. 10 cm = 1 km, and lengths show in real units (a 7 cm line reads 0.7 km).



Equation editor with symbol palette and live preview.

Equations: type LaTeX-style text such as `\frac{a}{b}`, `x^{2}`, `\sqrt{x}`, integrals, sums, limits, matrices and Greek letters – or tap the symbol palette. The preview updates as you type, and equations stay editable.



Function graph with parameter sliders.

Graphs: type a function like `2x + 1` or `a*sin(b x) + c`. Any letter other than x (a, b, c…) automatically gets its own **slider**, so you can show students how the graph changes. You can zoom, pan, place points on the curve, add vectors and make a table of values.

6. Pages, saving and exporting

- **Pages:** add, duplicate, clear or delete pages. The page navigator shows small pictures of all pages – drag to reorder, tap to open, rename.
- **Page size:** 16:9 board, 4:3, A4, A3 or custom size in cm. 1 cm on the page is always 1 cm in measurements.
- **Backgrounds / templates:** blackboard, whiteboard, grid, dots, ruled lines, graph paper, music staves, coordinate plane, or your own picture.
- **Saving:** lessons are saved as **.classboard** files.
- **Autosave & crash recovery:** it keeps a backup copy. If the computer suddenly shuts down, next time it offers to restore the lesson.
- **Import:** pictures, PDF files and PowerPoint (.ppt/.pptx) – every page or slide becomes a board page you can write on.
- **Export:** PDF (whole lesson or chosen pages), PowerPoint (one slide per page) or a PNG picture. Exports always contain the full page, whatever the zoom.

Handy keyboard shortcuts

Keys	Action
Ctrl+Z / Ctrl+Y	Undo / redo
Ctrl+S / Ctrl+O / Ctrl+N	Save / open / new lesson
Ctrl+C / Ctrl+V / Ctrl+D	Copy / paste / duplicate
P, E, V, T	Pen, eraser, select, text
Page Up / Page Down, Ctrl+M	Previous / next page, new page
F11	Full screen
Esc	Close pop-up / clear selection

7. How it’s built inside (simple version)

The program is split into small parts (**modules**). Each part has one job – like departments in a school. Only the main “app” part talks to all of them.

Part	Its job	School example
app	Main window; connects all other parts	Principal
input	Understands mouse, touch and pen; decides if a finger is writing, zooming or wiping	Gatekeeper
tools	Pen, eraser, select, shapes, text, measure, graph tools	Stationery box

Part	Its job	School example
document	Stores pages and everything on them; keeps the undo history	Notebook
canvas	Draws the page on screen quickly	Blackboard
geometry / math / graph	Instruments, measurements, equation drawing, function graphs	Maths lab
ai	Shape recogniser and handwriting-to-equation recogniser	Smart helper
storage / export	Saving, autosave, PDF import; PDF / PPTX / PNG export	Library
ui	Buttons, ribbon, pop-ups, colours (theme), icons	Decoration

A few smart design ideas

- **One undo list:** every change is saved as a "command" (add, remove, change…). That's why one Undo button works for every tool.
- **Fast drawing:** the screen keeps a ready-made picture and only redraws the small area that changed.
- **What you see is what you export:** the same drawing code is used for the screen, page thumbnails, PDF and PowerPoint.
- **Safe saving:** files are saved "atomically" – the old file is only replaced once the new one is fully written, so a power cut can't corrupt it.
- **Tested:** the project has automatic tests (tests folder) for shapes, geometry, saving, exporting, the equation maker and the UI.

Install and run

Easy way (Windows): download `ClassBoard-Setup-<version>-x64.exe` from the GitHub releases page and install it like any app. You get a Start menu entry, a desktop shortcut, and .classboard files open with a double-click.

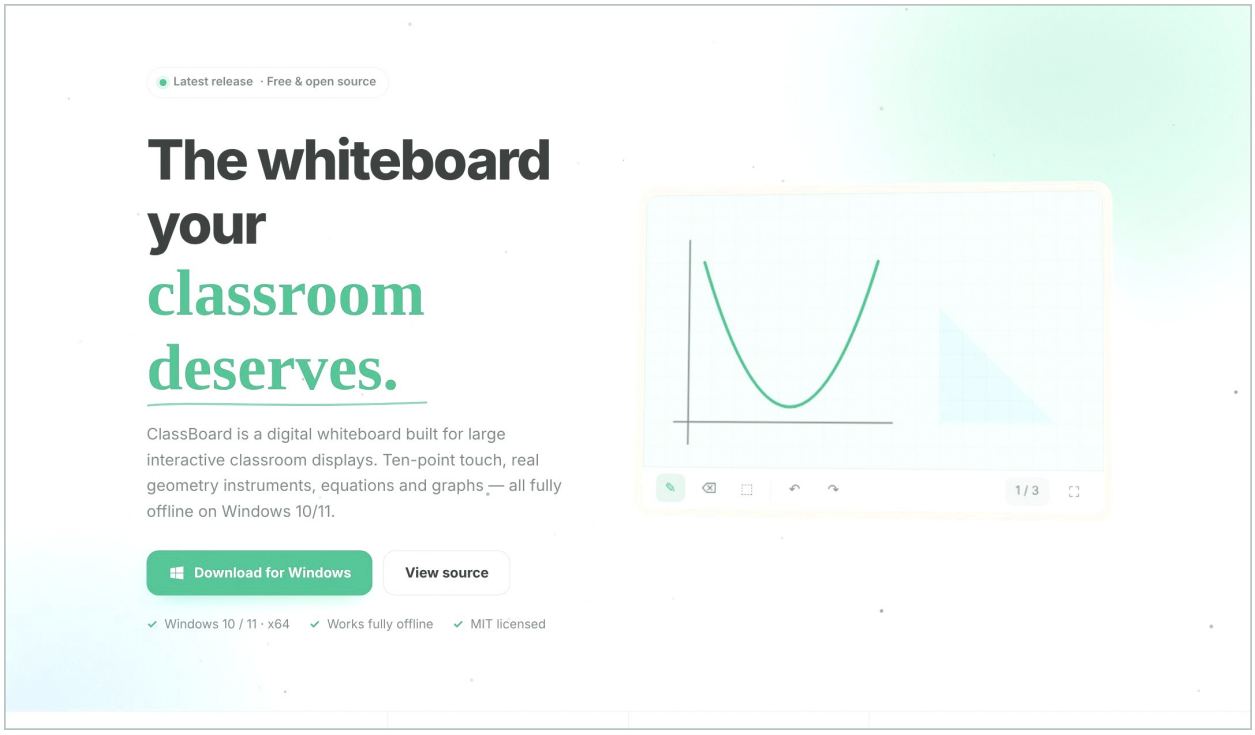
Build from code (needs CMake 3.16+, a C++17 compiler and Qt 5.15):

```
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=Release
cmake --build build
ctest --test-dir build --output-on-failure
./build/bin/ClassBoard
```

8. The ClassBoard website

ClassBoard has its own website where anyone can read about it, try small live demos and download the app. It's the easiest thing to show people first – just open the link on a phone or laptop.

Website: `an0nym0us-glitch.github.io/Smart-board-website`

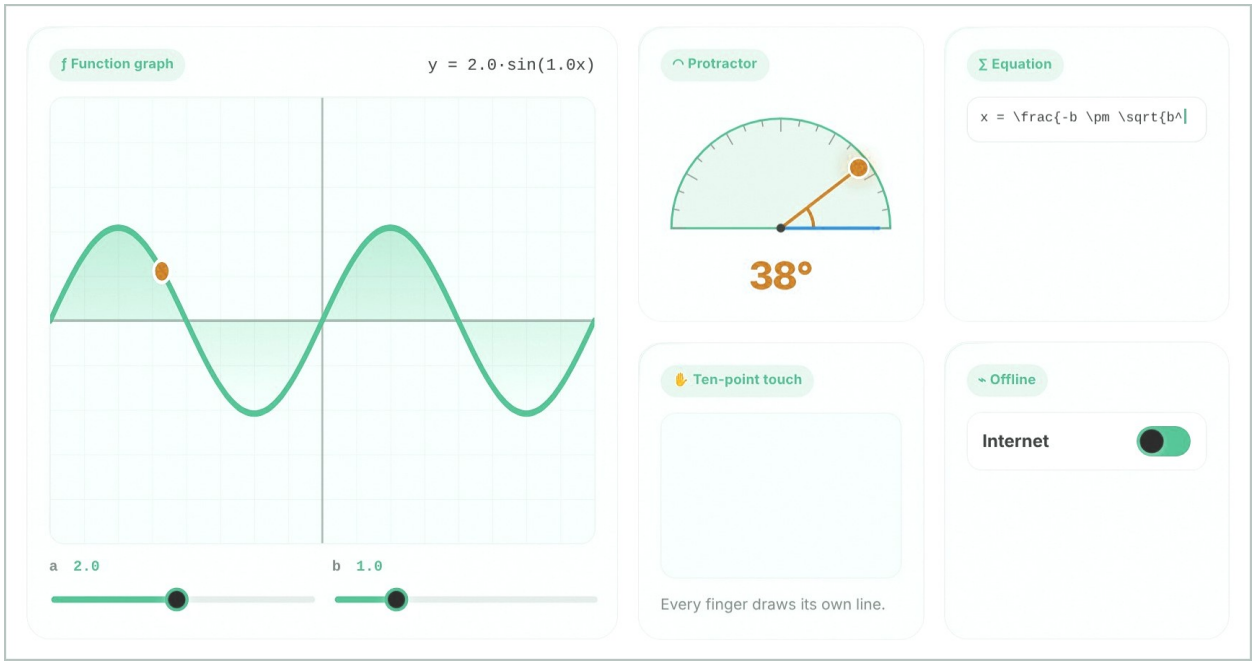


The top of the website: “The whiteboard your classroom deserves.” with a Download for Windows button.

What’s on the website

Section	What people will find there
Home (top)	Headline, short description, **Download for Windows** and **View source** buttons, and a small board you can draw on.
Features	12 cards – ink, Magic Highlighter, palm wipe, instruments, live geometry, equations, graphs, pages, import/export, autosave, text/images/tables, fully offline.
Demos	Little live demos you can play with right in the browser (see next page).
Tour	Real screenshots of the app in tabs: More menu, Pen, Instruments, Equation, Graph.
Touch	How ten-point touch, palm rejection and palm wipe work, plus keyboard shortcuts.
Download	The installer for Windows 10/11 in 3 steps: download → run → open from the Start menu.
Build	Commands for developers who want to build it from the source code.
FAQ	Common questions teachers ask – is it free, does it need internet, which screens work, etc.

Live demos – try it in the browser



The Demos section: a function graph with sliders, a protractor, an equation box, ten-point touch and an offline switch.

Demo	What to show people
Function graph	Move the **a** and **b** sliders and watch y = a·sin(bx) change – just like the graph tool in the app.
Protractor	Drag the orange knob and the angle updates live.
Equation	Type LaTeX like the quadratic formula and see it drawn as real maths.
Ten-point touch	On a phone or touch screen, put several fingers down – each draws its own line.
Offline	Switch the internet off – ClassBoard still works (no account, no cloud, no tracking).

Other handy website features

- **Search (Ctrl + K):** a quick command box to jump to any part of the page.
- **Light / dark switch:** the sun button changes between the dark blackboard theme and a whiteboard theme.
- **Always the newest version:** the download button checks GitHub for the latest release, so people always get the newest installer.
- **Works on phones:** the page adjusts to small screens, so you can show it from your mobile.

> **Say it like this:** *"Just open the ClassBoard website. You can read what it does, try the graph and protractor right in the browser, and download the free installer for Windows with one click."*

How to present it (2-minute plan)
1. Open the website and read the headline. Say it's free, open source and works offline.
2. Scroll to **Demos** – let them move the graph sliders and the protractor.
3. Open **Tour** – click the tabs to show real screenshots of the app.
4. End at **Download** – show how easy installing is (3 steps).

9. Questions people may ask you

Q. Is it really AI?

It's "smart" rather than a big AI like ChatGPT. It uses pattern matching and geometry to recognise shapes and handwriting. That makes it fast, offline, and predictable – it gives the same answer every time.

Q. Does it need internet?

No. Everything, including the smart features, runs on the computer itself.

Q. What if the equation maker reads it wrong?

It always shows a preview first. The teacher can Accept, Edit or Cancel. If it isn't sure, it says so instead of guessing. And one Undo brings the handwriting back.

Q. Which symbols can it read?

Digits 0–9, x, y, a, b, c, n, +, −, =, brackets, powers like x^2 and simple fractions.

Q. Can many students write together?

Yes – up to 10 fingers (or as many as the screen supports), each drawing its own line.

Q. How do you stop your hand from drawing?

When the pen is near the screen, finger touches are ignored.

Q. What happens if the computer crashes?

Autosave keeps a backup. Next time ClassBoard starts, it offers to restore the lesson.

Q. Can students get the notes?

Yes – export the lesson as PDF or PowerPoint and share it.

Q. Can I use my old PowerPoint or PDF?

Yes – import it, and each slide or page becomes a board page you can write on.

Q. What language is it written in?

C++17, using the Qt 5.15 framework for the window, touch and drawing.

Q. Where can I see it or download it?

On the ClassBoard website: an0nym0us-glitch.github.io/Smart-board-website – it has live demos and a Download for Windows button.

Q. Is it free?

Yes. It's open source under the MIT licence – anyone can use, study and change it.

10. Simple words – a mini glossary

If someone asks what a word means, here is a one-line answer.

Word	Simple meaning
Ribbon	The strip of main buttons along the bottom of the screen.
Popover	A small pop-up panel that opens right next to the button you tapped.
Offline	Works without internet – everything runs on the classroom computer.
Open source	The code is public on GitHub; anyone can read it, use it and improve it.
MIT licence	A very free licence: you can use, copy and change the software at no cost.
Ten-point touch	The screen can feel up to 10 fingers at the same time.
Palm rejection	Your hand resting on the screen is ignored while you write with the pen.
LaTeX	A short way to type maths, e.g. \frac{1}{2} means ½.
Recogniser	The part of the program that guesses what you drew (a shape or a maths symbol).
Vector PDF	A PDF that stays sharp at any zoom, with real text you can search.
Autosave	The board saves a backup copy by itself every few minutes.
.classboard file	The file type ClassBoard uses to save a whole lesson.

Remember – the main points
1. It's a **free digital blackboard** for classroom touch screens.
2. It has **smart tools**: neat shapes, handwriting → equation, graphs, geometry instruments.
3. It's **offline and safe**: autosave, one undo for everything, and export to PDF / PowerPoint.
4. Try it and download it from the **website**.

My notes